Ali Nasr

AI Systems Architect | R&D Engineer

ainlamyae@gmail.com | Website | GitHub | LinkedIn | Google Scholar

About

AI Systems Architect and R&D Engineer with extensive expertise in the design and MBSE of autonomous platforms, from agentic or intelligent decision-making to real-time robotic control.

Core Engineering Expertise

- Requirements Engineering, System & Component Architecture Design
- Prompt Engineering, Multi-Agent Orchestration, Reliability Pipelines, Human-in-the-Loop Validation
- Modeling, Identification, Training, Validation, and Simulation
- Software-in-the-loop Testing, Real-time Control Development, Signal Processing, Runtime Monitoring
- Embedded Software, Communication Systems, Sensor & Actuator Integration
- Calibration, Prototyping, and Optimization

Systems Experience

- **Generative AI:** Text Generation, Code Generation (5G Scripts & Test Cases), NL Interpretation, RAG, LLM-as-a-Judge
- **Intelligent Systems:** Autonomous Driving (Apollo Auto, End-to-End Interfuser)
- **Robotics:** Industrial, Serial & Parallel (Cable, Stewart), Humanoid, Biped, UGV, UAV (Quadcopter), Exoskeleton (Upper & Lower Limb)
- **Biological Systems:** Musculoskeletal Modeling, sEMG Analysis
- **Physical Systems:** Electrical, Hydraulic, Pneumatic, Mechanical

Skills

- **Generative AI:** LangChain, Ollama
- **Machine Learning:** ANN, RNN, CNN, RCNN, LSTM, GRU, SVM
- **Programming:** Python (NumPy, SciPy, Matplotlib, Pandas, scikit-learn), MATLAB, C++
- **System Modeling & Simulation:** MATLAB (Simulink, Simscape Multibody, Deep Learning Toolbox, Neural Networks, GA, PSO), MapleSim, LabVIEW
- **Embedded Systems:** Raspberry Pi, NVIDIA Jetson, Arduino, AVR Microcontrollers, PLCs (Siemens, Delta, Vigor)
- **Control Methods:** PID, LQR, LQG, CTM, MPC, NMPC, Feedback Linearization, Impedance, Adaptive, Sliding Mode, Fuzzy Logic, ANFIS
- **CAD & Mechanical Design:** CATIA, SolidWorks
- **Hands-on Skills:** Mechanical Fabrication, Soldering, 3D Printing

Experience

Senior R&D Engineer (GenAI System, Assurance Architect, and Design) Nov 2023 – Present
Huawei Technologies Canada — Waterloo, Ontario, Canada
Permanent Full-time
Trustworthy Technology and Engineering Laboratory, Waterloo Complex Intelligent Systems Laboratory
LLMGuard Wrapper: A Runtime Assurance Framework for Reliable LLM Systems

- Designed and delivered a runtime monitoring and assurance architecture (GuardRailClient) for LLM-

based multi-agent systems, enabling reliable and production-ready AI behavior through integrated monitoring, validation, and correction
- Introduced a novel confidence-driven methodology that leverages token-level LogProbs aggregated into JSON-value scores to detect uncertainty and autonomously identify erroneous outputs without labeled data
- Developed a modular, scalable framework (wrapper-based integration) that seamlessly plugs into existing LLM pipelines with minimal code changes, supporting both synchronous and asynchronous execution
- Implemented an adaptive validation and recovery pipeline using LLM-as-a-Judge, multi-strategy correction (prompting, model switching, self-consistency), and optional human-in-the-loop escalation for high-risk cases
- Engineered an efficient field-level correction mechanism that selectively fixes low-confidence outputs, reducing recomputation, lowering latency, and improving system stability in real-world deployments
- Demonstrated measurable reliability improvements in a script generation system, successfully detecting ambiguous/erroneous outputs and enabling automated correction under real-world and adversarial scenarios

Uncertainty in LLMs: Theory, Methods, and Practical Reliability Frameworks

- Conducted a technical review and synthesis of uncertainty in LLMs, covering theory, state-of-the-art methods, and practical applications for improving reliability in real-world AI systems
- Analyzed and formalized key uncertainty concepts (aleatoric, epistemic, semantic, entropy-based metrics), translating research insights into actionable frameworks for inference-time risk assessment
- Evaluated and compared multiple uncertainty estimation techniques (token probability, entropy, sampling, self-check, and ensemble methods), identifying strengths, limitations, and failure modes such as "confidently wrong" predictions
- Designed and executed experiments on hallucination detection using uncertainty features, including entropy, token probabilities, and Dirichlet-based metrics, demonstrating effective model-agnostic detection without ground truth
- Developed and validated a multi-feature classification approach for hallucination detection, showing improved performance over single-metric methods and highlighting the importance of combining uncertainty signals
- Derived system-level insights for production AI systems, including the impact of temperature on uncertainty, limitations of entropy-based metrics, and the need for multi-step uncertainty tracking in agent-based workflows

Explainable & Reliable System Architecture for LLM-based Script Generator

- Architected a multi-stage LLM reliability pipeline with confidence estimation, output gating, and policy-driven decision enforcement
- Implemented probability-based and token-level uncertainty scoring to dynamically control acceptance, remediation, or escalation of LLM outputs
- Applied voting and consensus mechanisms across multiple LLM outputs to improve consistency and select highest-agreement results
- Engineered an assurance loop with temperature adjustment, prompt perturbation, and model switching for iterative refinement
- Implemented a Decision Maker using a Markov Decision Process (MDP) for policy enforcement under system constraints
- Enabled fail-safe escalation paths including re-execution with alternative models, LLM-as-Judge evaluation, and human-in-the-loop review
- Designed an explainable JSON-based output schema ensuring traceability of generated variables and intermediate decisions
- Proposed metamorphic prompt perturbation via semantic-preserving sentence swapping (Chinese inputs) to test robustness and consistency

Multi-Agent Foundation Model-based AI for Autonomous Software Testing Platform

- Designed a multi-agent system architecture leveraging foundation model-based AI (RAG-based LLMs) for a fully autonomous software testing platform
- Developed run-time accuracy and consistency monitoring mechanisms for collaborative AI agents
- Collaborated on embodied AI concepts for general-purpose intelligent assistance systems

2D Bird's-Eye View Risk Monitoring for Autonomous Driving

- Developed a CNN-based risk assessment module that processes 2D bird's-eye views of the vehicle's surroundings to predict the future driving risk
- Implemented risk detection using a 2D Time-to-Collision (TTC) algorithm for precived object and obstructed phantom object
- Coded and integrated C++ modules into Baidu Apollo Auto ADS, enabling real-time risk evaluation and decision-making
- Verified and tested the system under diverse conditions, analyzing accident scenarios and validating robustness of the risk detection algorithms
- Animated 2D bird's-eye views and reconstructed accident cases from China for kinematic analysis and validation

Game Theory Orchestration of Multi-Agent LLM-based Reinforcement Learning

- Proposed and scoped a game-theoretic orchestration framework for multi-agent LLM-based reinforcement learning systems

Uncertainty Monitoring System for LLM-Based Generative Software

- Implemented the core uncertainty computation for the LLM outputs, forming the foundation for the monitoring system
- Developed the decision-making logic to act on uncertainty measures and partially contributed to coding the overall system
- Assisted in project planning and integration, supporting the application of uncertainty metrics to JSON-based outputs

Run-time Behavioural Validation of Self-adaptive AI-enabled Systems

- Led and coordinated the uOttawa engineering team, aligning technical efforts and ensuring milestone delivery
- Extended metamorphic testing methodologies to LLM-based software and defined practical application strategies
- Collected and formalized metamorphic relations for LLM applications to detect subtle undesirable behaviors beyond explicit failures
- Supervised development of a search-based test generation approach for autonomous driving scenarios
- Oversaw case studies implemented in CARLA simulator on the Interfuser autonomous driving system
- Directed automated offline system-level testing combining Metamorphic Testing with Evolutionary Algorithms

LLM-based System Reliability

- Guided implementation of probability extraction from LLMs within LangChain

C++ Real-Time Processing Engine for Monitoring Software

- Developed prototype of C++ modules for processing as part of the monitoring software project
- Conducted testing and validation of the C++ components to ensure reliability before integration
- Delivered ready-to-use modules to assistant for integration into the larger framework

Strategic Insights and Innovations from Global Technology Conferences

- 14th International Conference on Learning Representations (ICLR), Rio de Janeiro, Brazil, Apr 2026
- 39th Annual Conference on Neural Information Processing Systems (NeurIPS), Mexico City, Mexico, Nov–Dec 2025
- 34th International Joint Conference on Artificial Intelligence (IJCAI), Montreal, Canada, Aug 2025
- 38th Annual Conference on Neural Information Processing Systems (NeurIPS), Vancouver, Canada, Dec 2024
- OpenMind Research Institute Banff Retreat, Banff, Canada, Oct 2024
- IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Seattle, USA, Jun 2024
- Systems Engineering Workshop for Complex Intelligent Systems, Hong Kong Science Park, Hong Kong, May 2024
- Symposium on Innovative Robotics Technologies and Applications, Hong Kong Science Park, Hong Kong, May 2024
- Strategy and Technology Workshop by Huawei R&D Teams, Shenzhen and Dongguan, China, May 2024
- 18th Annual IEEE International Systems Conference (SysCon), Montreal, Canada, Apr 2024
- 38th Annual AAAI Conference on Artificial Intelligence (AAAI), Vancouver, Canada, Feb 2024

Project Manager & Technical Interviewer Responsibilities

- Managed project workflows using Jira, tracked progress, led status updates, and prepared detailed project progress reports
- Identified team skill gaps and collaborated with leadership to define role requirements, and draft job descriptions
- Screened 1,000+ resumes, and shortlisted high-potential candidates based on technical qualifications
- Conducted 50+ remote technical interviews, evaluating problem-solving and programming ability

Senior R&D Engineer (AI Assurance in Autonomous Driving) Jun 2023 – Nov 2023
Huawei Technologies Canada — Waterloo, Ontario, Canada
Contract Full-time
Trustworthy Technology and Engineering Laboratory, Waterloo Complex Intelligent Systems Laboratory
Run-time Assurance (AI Consistency and Accuracy) System for Autonomous Driving Systems

- Researched failure measurement and monitoring of learning-enabled control systems across Perception, Prediction, and Planning modules
- Performed systems-theoretic process analysis and defined risk boundaries for autonomous driving systems
- Formulated and implemented C++ algorithms for monitoring, risk assessment, and secure disengagement protocols
- Designed a comprehensive run-time assurance architecture for AI consistency and accuracy validation
- Evaluated advanced autonomous driving modules by integrating Baidu Apollo Auto with CARLA Simulator using C++ and Python
- Designed pre-crash, adverse weather, and empirical simulation scenarios to validate robustness and safety

Run-time Assurance (Vehicle Safety) System for Autonomous Driving Systems

- Designed, developed, and tested vehicle recovery software for safety violations in autonomous driving systems
- Implemented proactive risk prediction and adaptive threshold tuning mechanisms
- Extracted requirements from ISO 26262, ISO 25058, and AVSC 6202103 to design compliant safety-oriented software architecture
- Developed AI system testing strategies including white-box and black-box evaluation for behavioral and activation coverage analysis

- Prototyped system components in Python and C++ and collaborated with simulation and production teams for validation

Postdoctoral Research Associate (Autonomous Control) Jan 2023 – Mar 2024
University of Waterloo — Waterloo, Ontario, Canada
Contract Full-time
Department of Systems Design Engineering *Real-time Embedded Controller within Different Sensors and Intelligence Control*

- Successfully designed and implemented platform real-time communication and control software on Raspberry Pi 4B (Linux environment) to support advanced application features, resulting in a 20% increase in user engagement and a 30% reduction in software crashes
- Developed machine learning-based real-time control unit with Python (multi threaded and object-oriented), resulting in improved system performance and accuracy
- Proficient in developing mid-level programs with Bluetooth, CAN-bus, and SPI connections to achieve seamless communication between hardware and software
- Conducted rigorous feature validation using hardware-in-the-loop (HIL) testing, ensuring the robustness and reliability of the system

Electrical Energy Regenerative Shoulder Exoskeletons

- Implementation of energy-regenerating actuators in robotic exoskeletons to enhance efficiency and extend battery life by converting a portion of energy typically lost during negative mechanical work into electrical energy.
- Evaluated the electrical energy regeneration capabilities of a shoulder wearable exoskeleton.
- Motor parameters were determined through experiments using a joint dynamometer system.
- Developed an electro-mechanical model for human-exoskeleton interaction focusing on energy regeneration.
- Conducted task-based experiments to measure energy regeneration during real-life usage of the exoskeleton.
- Results compared with practical usage scenarios to assess feasibility; electricity generated was relatively low due to motor characteristics, but indicated potential for energy regeneration with certain limitations.

Human3.6M+: Generated Human Motion Data to Support 3D Pose, Motion, Kinetic, and Muscle Estimation

- Developed marker-less motion capture technology for 3D human pose estimation to overcome usability constraints of traditional marker-based systems.
- Synthesized dynamics and muscle properties from existing motion capture datasets to generate training and validation data for deep learning models.
- Reconstructed motions using kinematics, dynamics, and muscle modeling while ensuring adherence to International Society of Biomechanics standards.
- Integrated musculoskeletal model with customizable anthropometric features (height, mass, athleticism level, handedness, skin temperature) to enhance realism.
- Applied methodology to Human3.6M dataset, demonstrating kinematic similarity to ground truth markers and dynamic plausibility compared to experimental data.

Postdoctoral Responsibilities

- Organized a remote international workshop with Prof. Daniel J. Rixen's group from Technical University of Munich (TUM)
- Coordinated presentations on topics ranging from humanoid robot demonstrations to advanced research in biomechanics and control systems

Myography-based Controlled 1-DoF Hand Prosthesis

- Designed an electrically controlled single degree-of-freedom (1-DoF) forearm-hand prosthesis to aid pediatric amputees after Mine Blast Injuries in Ukraine.
- Achieved precise thumb controllability through electrical component development, control system refinement, mechanical design, and final assembly.
- Prototype serves as an active pediatric prosthesis using EMG-based control rather than muscle strength, despite being limited to a single finger.
- Future improvements planned: refine 3D model, optimize electronic housing, enhance EMG control precision, and incorporate motors for full-finger control.
- Prototype is a pivotal tool for initiating prosthesis control learning in rehabilitation.

MyoBuzz: Real-time Muscle Activation Monitoring in Dynamometry Experiments

- Developed MyoBuzz, a limb-worn EMG device providing real-time feedback on muscle activation for dynamometry experiments.
- Limb-specific band with strategically positioned electrodes, visual and auditory alerts, customizable activation threshold, and adjustable sensitivity.
- Electrodes are attachable via clips; strap ensures comfort; rechargeable 3.7V battery supports one-hour operation and two-hour recharge.
- Facilitates real-time monitoring and mitigation of inadvertent muscle activation during joint torque evaluations.

Human Motion Animation Using 3Ds Max for 3D Pose Estimation Database

- Guided in setting up 3Ds Max environment and workflow for creating human motion animations for synthetic dataset generation
- Provided joint position data and guidance to support annotation for training the 3D pose estimation machine learning model

Graduate Teaching Assistant Jan 2019 – Apr 2023
University of Waterloo — Waterloo, Ontario, Canada
Contract Part-time
Department of Systems Design Engineering *Teaching Responsibilities*

- Delivered tutorials and undergraduate instructional support
- Explained complex engineering concepts during laboratory sessions, office hours, and via email communication
- Evaluated assignments, projects, and examinations to assess student learning outcomes

Course List **Graduate Research Assistant (BioMechatronic Systems)** Sep 2018 – Dec 2022

University of Waterloo — Waterloo, Ontario, Canada
Contract Full-time
Motion Research Group *Design, Dynamics, and Control of Active-Passive Exoskeletons Robotic*

- Developed and verified adjustable biomechanical 20-DoF model using MATLAB and MapleSim
- Designed, optimized, and developed actuator of wearable robot in accordance with ISB standards
- Conducted research and developed predictive models using ANN, RNN, CNN, and RCNN to estimate biomechanical (kinematic and kinetic) variables from noisy electromyography sensor data
- Developed novel assist-as-needed control regulations for wearable robots using model-based predictive control and fuzzy logic techniques
- Implemented torque and motion real-time control of active wearable robots with sEMG sensors and CAN-bus connection, leading to a 15% improvement in accuracy and a 25% increase in robot speed
- Designed and implemented safety measures for autonomous systems to ensure compliance with human testing, industry regulations, and standards, resulting in successful attainment of safety approval and ethics certificate

Active-passive Shoulder Exoskeleton

- Developed active-passive shoulder exoskeleton integrating passive mechanisms and powered actuators to optimize human movement assistance.
- Established multibody kinematics and dynamics model for upper body, implemented optimal controller for reducing muscular activation torque.
- Optimized exoskeleton design to enhance user comfort and reduce fatigue while maintaining functional assistance.
- Results demonstrate substantial reduction in fatigue and functional improvement over fully passive or fully active systems.

MuscleNET: Biomechatronic Electromyography Signal Processing

- Introduced MuscleNET, a set of ML models (ANN, RNN, CNN, RCNN) mapping sEMG signals to joint angle, velocity, acceleration, torque, and activation torque.
- RobustMuscleNET RNN predicts control-oriented signals in real-time with 92% regression accuracy for joint angle and 86.5% for assistive torque.
- InverseMuscleNET resolves redundancy in inverse muscle models using RNNs, achieving 88–91% normalized regression between estimated and experimental muscle activation.
- Enables fast, direct estimation for real-time applications in rehabilitation and sports evaluation without invasive electrodes.

Scalable Human Musculoskeletal Model

- Developed a MapleSim biomechanics-based musculoskeletal (MSK) model covering upper and lower body movements for biomechanical analysis.
- Estimated biomechanical joint torque using muscle torque generators with active and passive components.
- Created adjustable multibody kinematic model accommodating diverse anthropometric measurements (sex, age, body mass, height, dominant side, physical activity, skin temperature) with 40 DoF.
- Validated model equations through simulations of joint range of motion and torque, demonstrating robust agreement with existing research findings.

Conferences & Academic Activities

- 18th International Symposium on Computer Methods in Biomechanics and Biomedical Engineering, Paris, France, May 2023
- Waterloo Robotics Day, Waterloo, Canada, March 2023
- 6th International Conference on Multibody System Dynamics, New Delhi, India, October 2022
- CRE-MSD Conference on Manual Handling, Mississauga, Canada, October 2022
- 5th North American Congress on Biomechanics, Ottawa, Canada, August 2022
- SYDE Graduate Symposium, Waterloo, Canada, June 2022
- 45th American Society of Biomechanics Annual Conference, Virtual, August 2021
- 45th Mechanisms and Robotics Conference, Virtual, August 2021
- 17th International Conference on Multibody Systems, Nonlinear Dynamics, and Control, Virtual, August 2021
- 7th International Conference of Control, Dynamic Systems, and Robotics, Virtual, November 2020
- 16th Annual Ontario Biomechanics Conference, Nottawasaga, Canada, March 2019

R&D Mid-Level Engineer (Robotics) Jan 2016 – Aug 2018
Techno Niro — Dorcheh, Esfahan, Iran
Contract Full-time Spatial 6 DoF 8 Cable-Driven Parallel Robot

- Conducted comprehensive inverse kinematic and dynamic modeling and analysis for the Spatial 6 DoF 8 Cable-Driven Parallel Robot.
- Validated the developed models using the SimMechanics Toolbox.
- Designed an optimal cable structure attachment for the robot.

- Simulated PD Partial, LQR, Robust, and Adaptive Control in the presence of noise and disturbance.

3 DoF Wrist Rehabilitation Robot

- Designed conceptual and mechanical parts for the 3 DoF Wrist Rehabilitation Robot and conducted kinematic and dynamic modeling.
- Implemented modified impedance control for enhanced human-robot interaction.
- Prototyped the wrist rehabilitation robot (Jan 2016 – Jan 2017), integrating modeling, mechanical design, and control for testing.

Graduate Marker Sep 2015 – Nov 2015
K. N. Toosi University of Technology — Tehran, Tehran, Iran
Remote
Faculty of Mechanical Engineering *Teaching Responsibilities*

- Assessed student learning through evaluating assignments, projects, and exams

Course List **Graduate Research Assistant (System Control)** Sep 2014 – Dec 2015

K. N. Toosi University of Technology — Tehran, Tehran, Iran
Contract Part-time
Advanced Robotics & Automated Systems (ARAS) Laboratory *Multiple Impedance Control of Hybrid Parallel-Serial Manipulator for Object Manipulation*

- Designed and developed a MATLAB/Simulink model for an 8 cable-driven parallel robot prototype with a serial manipulator, and verified its performance
- Conducted a comprehensive analysis of stiffness, sensitivity, singularity, and workspace volume to optimize the structure of a constraint and suspended parallel robot, using Genetic Algorithm (GA)
- Simulated and evaluated multiple impedance control algorithms, including serial and parallel, for object manipulation and human-robot force/moment interaction
- Prototyped the 8 cable constraint parallel robot, integrated image processing (vision) and IMU modules for precise position and orientation feedback, proposed an algorithm to address sensor shortage, and successfully implemented real-time motion control

Optimal Redundancy Resolution Scheme in a Cable-Driven Parallel Robot

- Developed an enhanced kinematic modeling of pulleys and cables for more accurate representation of cable-driven parallel robots
- Improved tension distribution analysis by integrating precise cable attachement modeling into the system framework
- Implemented a basic optimization approach using fmincon for estimating actuator force distribution

Sensory Feedback and Experimental Evaluation for Cable Robots with Uncertain Jacobian

- Led the mathematical modeling, algorithm coding, and manufacturing of the prototype of the cable-driven parallel robot
- Integrated the experimental and theoretical foundations on the robot's design and optimization approach
- Assisted the team with hardware, software, and modeling efforts

Visual Servoing in a Cable Robot Using Microsoft Kinect V2 Sensor

- Developed the kinematic model and control code for the parallel cable robot
- Supported integration between the robot and the visual tracking system
- Guided the use and modification of the Microsoft Kinect sensor for experiments

Digital Image Processing of Ball-and-Plate System Using Artificial Neural Network

- Integrated camera with MATLAB for real-time ball position tracking using classical image processing

techniques
- Contributed to development of a vision-based closed-loop control system
- Improved stability and responsiveness of ball positioning on a two-axis tilting platform

R&D Junior Engineer (Robotics) Sep 2013 – Dec 2015
Techno Niro — Dorcheh, Esfahan, Iran
Contract Part-time 6 DoF Industrial Serial Manipulator Modeling & Control

- Conducted comprehensive kinematic and dynamic modeling and analysis for the 6 DoF Industrial Serial Manipulator, assessing workspace, dexterity, and performance
- Utilized CATIA software for the design of mechanical parts, performed finite element analysis, and supervised the manufacturing process using CNC machines
- Prepared the hardware-in-loop of the control system using LabVIEW and MATLAB, and conducted motor parameters identification
- Implemented PID control, adjusted control coefficients for different maneuvers, and planned trajectory using ARM-controller programming

R&D Intern (Automation) Jun 2013 – Aug 2013
Kian Control Apadana Co. (Festo Authorized Distributor) — Isfahan, Esfahan, Iran
Internship Automation and Mechatronic System Development

- Developed the mechatronic scheme for a steel plant using ePLAN software, including designing the sensors, actuators, and wiring for the system
- Conducted extensive research on the hydraulic variable displacement pump action, exploring various modes of operation and optimizing their efficiency

Undergraduate Marker Jan 2013 – May 2013
Isfahan University of Technology — Isfahan, Esfahan, Iran
Hybrid Teaching Responsibilities

- Assessed student learning by marking assignments

Course List **Undergraduate Research Assistant** Sep 2012 – Aug 2013

Isfahan University of Technology — Isfahan, Esfahan, Iran
Contract Part-time
Department of Mechanical Engineering *Joint Space Energy Optimization of Tethered Space Robot for End-Effector Optimal Path*

- Conducted research on tethered satellite systems and tether-connected space manipulators for applications such as satellite repair and space debris removal.
- Modeled a tethered space manipulator system using a straight, rigid rod for the tether, restricting motion to the orbital plane with the platform following a circular orbit around Earth.
- Analyzed system dynamics with the end-effector tracing prescribed trajectories, monitoring tether tension to prevent compressive forces.
- Investigated feasible end-effector trajectories under torque restrictions, ensuring torques on tether attachment points were zero.
- Proposed a method to determine possible end-effector paths between two points while satisfying the zero-torque constraint on the tether.
- Developed and simulated a nonlinear control algorithm for vibration reduction in a redundant robot by adjusting cable length, validated through simulations and experiments.
- Performed comprehensive modeling of forward kinematics, workspace, and dynamics of a 4-DoF tethered space robot to evaluate performance and energy efficiency.

R&D Intern Jun 2009 – Aug 2013
Techno Niro — Dorcheh, Esfahan, Iran

Contract Part-time Plant Mechatronic Design, Maintenance, and Project Management

- Developed and supervised the implementation of a comprehensive mechatronic scheme for the plant, utilizing ePlan software and programming Siemens PLC for optimal performance
- Provided proactive maintenance and effective troubleshooting for mechanical systems and equipment
- Actively participated in project planning and execution, fostering collaboration to meet project deadlines
- Ensured adherence to quality standards, conducted regular inspections, and maintained accurate documentation of all mechanical systems
- Stayed abreast of industry trends and advancements, and pursued continuous learning through workshops and training

Education

Doctor of Philosophy (Ph.D.) in Systems Design Engineering Sep 2018 – Dec 2022
University of Waterloo — Waterloo, Ontario, Canada
Major: Intelligent & Automated Systems; Biomechatronics Systems

Master of Science (M.Sc.) in Mechatronics Engineering Sep 2013 – Dec 2015
K. N. Toosi University of Technology — Tehran, Tehran, Iran
Major: Design Robotic and Mechatronic Systems

Bachelor of Science (B.Sc.) in Mechanical Engineering Sep 2009 – Aug 2013
Isfahan University of Technology — Isfahan, Esfahan, Iran
Major: Mechatronics and Dynamic Systems

Publications

Journal (18)

[J18] H. Yousefizadeh, S. Gu, L. Briand, and **A. Nasr**, "Constrained co-evolutionary metamorphic differential testing for autonomous systems with an interpretability approach", *ACM Transactions on Software Engineering and Methodology (TOSEM)*, 2026.

[J17] H. Yousefizadeh, S. Gu, L. Briand, and **A. Nasr**, "Using cooperative co-evolutionary search to generate metamorphic test cases for autonomous driving systems", *IEEE Transactions on Software Engineering*, vol. 15, no. 6, pp. 1882-1911, 2025.

[J16] **A. Nasr** and J. McPhee, "Harnessing biomechanical energy using shoulder exoskeleton: Electrical energy regeneration", *ASME Journal of Mechanisms and Robotics*, vol. 17, no. 9, p. 091005, 2025.

[J15] **A. Nasr**, K. Inkol, and J. McPhee, "Safety in wearable robotic exoskeletons: Design, control, and testing guidelines", *ASME Journal of Mechanisms and Robotics*, vol. 17, no. 5, p. 050801, 2025.

[J14] **A. Nasr**, K. Zhu, and J. McPhee, "Using musculoskeletal models to generate physically-consistent data for 3D human pose, kinematic, dynamic, and muscle estimation", *Springer Journal of Multibody System Dynamics*, vol. 64, pp. 737-770, 2024.

[J13] S. Bell, **A. Nasr**, and J. McPhee, "General muscle torque generator model for a two degree-of-freedom shoulder joint", *ASME Journal of Biomechanical Engineering*, vol. 146, no. 8, p. 081008, 2024.

[J12] **A. Nasr** and J. McPhee, "Scalable musculoskeletal model for dynamic simulations of lower body movement", *Taylor & Francis Journal of Computer Methods in Biomechanics and Biomedical Engineering*, vol. 28, no. 8, pp. 1196-1222, 2024.

[J11] **A. Nasr**, C. Dickerson, and J. McPhee, "Experimental study of fully passive, fully active, and active-passive upper-limb exoskeleton efficiency: An assessment of lifting tasks", *MDPI Journal of Sensors*, vol. 24, no. 1, p. 63, 2023.

[J10] **A. Nasr**, S. Bell, R. Whittaker, C. Dickerson, and J. McPhee, "Robust machine learning mapping of sEMG signals to future actuator commands in biomechatronic devices", *Springer Journal of Bionic Engineering*, vol. 21, pp. 270-287, 2023.

[J9] **A. Nasr**, A. Hashemi, and J. McPhee, "Scalable musculoskeletal model for dynamic simulations of upper body movement", *Taylor & Francis Journal of Computer Methods in Biomechanics and Biomedical Engineering*, vol. 27, no. 3, pp. 306-337, 2023.

[J8] **A. Nasr**, J. Hunter, C. Dickerson, and J. McPhee, "Evaluation of a machine-learning-driven active-passive upper-limb exoskeleton robot: Experimental human-in-the-loop study", *Cambridge University Press Journal of Wearable Technologies*, vol. 4, p. e13, 2023.

[J7] **A. Nasr**, S. Bell, and J. McPhee, "Optimal design of active-passive shoulder exoskeletons: A computational modeling

of human-robot interaction", *Springer Journal of Multibody System Dynamics*, vol. 57, pp. 73-106, 2022.

[J6] M. Febrer-Nafria, **A. Nasr**, M. Ezati, P. Brown, J. Font-Llagunes, and J. McPhee, "Predictive multibody dynamic simulation of human neuromusculoskeletal systems: A review", *Springer Journal of Multibody System Dynamics*, vol. 58, pp. 299-339, 2022.

[J5] **A. Nasr**, S. Ferguson, and J. McPhee, "Model-based design and optimization of passive shoulder exoskeletons", *ASME Journal of Computational and Nonlinear Dynamics*, vol. 17, no. 5, p. 051004, 2022.

[J4] **A. Nasr**, A. Hashemi, and J. McPhee, "Model-based mid-level regulation for assist-as-needed hierarchical control of wearable robots: A computational study of human-robot adaptation", *MDPI Journal of Robotics*, vol. 11, no. 1, p. 20, 2022.

[J3] **A. Nasr**, K. Inkol, S. Bell, and J. McPhee, "InverseMuscleNET: alternative machine learning solution to static optimization and inverse muscle modeling", *Frontiers Media SA Journal of Frontiers in Computational Neuroscience*, vol. 15, p. 759489, 2021.

[J2] **A. Nasr**, S. Bell, J. He, R. Whittaker, C. Dickerson, N. Jiang, and J. McPhee, "MuscleNET: Mapping electromyography to kinematic and dynamic biomechanical variables by machine learning", *IOP Journal of Neural Engineering*, vol. 18, no. 4, p. 0460d3, 2021.

[J1] **A. Nasr** and S. Moosavian, "Multi-objective optimization design of spatial cable-driven parallel robot equipped with a serial manipulator", *Tarbiat Modares University Journal of Modares Mechanical Engineering*, vol. 16, no. 1, pp. 29-40, 2016.

Conference (23)

[C23] H. Yousefizadeh, S. Gu, L. Briand, and **A. Nasr**, "Using cooperative co-evolutionary search to generate metamorphic test cases for autonomous driving systems", *Proceedings of the IEEE/ACM International Conference on Software Engineering (ICSE)*, Rio de Janeiro, Brazil, 2026.

[C22] K. Zhu, **A. Nasr**, A. Wong, and J. McPhee, "3D human pose and torque estimation from monocular video", *Proceedings of the IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR)*, Seattle, WA, USA, 2024.

[C21] **A. Nasr** and J. McPhee, "Inverse kinematic, dynamic, and muscle simulation from video files: Toward markerless 3D human pose, torque, and muscle estimation", *Proceedings of the 7th International Conference on Multibody System Dynamics*, Madison, Wisconsin, USA, 2024.

[C20] V. Korovkina, **A. Nasr**, and J. McPhee, "Optimal EMG placement for hand prosthetic control", *Proceedings of the 18th Ontario Biomechanics Conference (OBC)*, Toronto, ON, Canada, 2024.

[C19] N. Haraguchi, **A. Nasr**, K. Inkol, K. Hase, and J. McPhee, "Human and passive lower-limb exoskeleton interaction analysis: Computational study with dynamics simulation using nonlinear model predictive control", *IEEE Proceedings of the 62nd Annual Conference of the Society of Instrument and Control Engineers (SICE)*, Tsu, Japan, 2023.

[C18] J. McPhee and **A. Nasr**, "Multibody system dynamics: A fundamental tool for biomechatronic system design", *Proceedings of the 12th ECCOMAS Thematic Conference on Multibody Dynamics*, Lisbon, Portugal, 2023.

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[C7] M. Mousavi, M. Ghanbari, **A. Nasr**, S. Moosavian, and P. Zarafshan, "Sensory feedback performance improvement on robocab: An experimental approach to wire-driven parallel manipulator", *IEEE Proceedings of the 4th International Conference on Robotics and Mechatronics (ICRoM)*, Tehran, Iran, pp. 33-38, 2016.

[C6] M. Nabipour, N. Arteghzadeh, S. Moosavian, and **A. Nasr**, "Visual servoing in a cable robot using Microsoft Kinect v2 sensor", *IEEE Proceedings of the 4th International Conference on Robotics and Mechatronics (ICRoM)*, Tehran, Iran, pp. 560-565, 2016.

[C5] **A. Nasr** and S. Moosavian, "Multi-criteria design of 6-dof fully-constrained cable driven redundant parallel manipulator", *IEEE Proceedings of the 3rd International Conference on Robotics and Mechatronics (ICRoM)*, Tehran, Iran, pp. 001-006, 2015.

[C4] M. Sajadi, **A. Nasr**, S. Moosavian, and H. Zohoor, "Mechanical design, fabrication, kinematics and dynamics modeling, multiple impedance control of a wrist rehabilitation robot", *IEEE Proceedings of the 3rd International Conference on Robotics and Mechatronics (ICRoM)*, Tehran, Iran, pp. 290-295, 2015.

[C3] Y. Parandian, H. Arabshahi, **A. Nasr**, and S. Moosavian, "Time optimized digital image processing of ball and plate system using artificial neural network", *IEEE Proceedings of the 3rd International Conference on Robotics and Mechatronics (ICRoM)*, Tehran, Iran, pp. 146-151, 2015.

[C2] **A. Nasr**, F. Asadi, K. Azami, and S. Ziaei-Rad, "Optimization of trunk angle in biped robots in generating a stable path in up-stair walking", *Iranian Society of Mechanical Engineers (ISME) Proceedings of the 23rd Annual Conference of International Society of Mechanical Engineers*, Tehran, Iran, p. HN10114881786, 2015.

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Patent (1)

[P1] **A. Nasr** and F. Risacher, "Method and a system for run-time operational monitoring of an autonomous vehicle", *U.S. Patent*, 2024.

Thesis (3)

[T3] **A. Nasr**, "Design, Dynamics, and Control of Active-passive Upper-limb Exoskeleton Robots", Doctor of Philosophy, *University of Waterloo*, 2022.

[T2] **A. Nasr**, "Multiple Impedance Control of Hybrid Parallel-Serial Manipulator for Object Manipulation", Master of Applied Science, *K.N. Toosi University of Technology*, 2015.

[T1] **A. Nasr**, "Optimizing Joint Space Energy of Redundant Tethered Space Robots", Bachelor of Applied Science, *Isfahan University of Technology*, 2013.

Projects

- **AI-Powered Exam Preparation & Assessment Platform (Full-Stack Distributed SaaS Architecture)** May 2026
  - Architected an AI-powered examination & assessment platform using Next.js 15, React 18, TypeScript, Node.js, Express.js, PostgreSQL, Firebase Auth, Stripe, and Docker.
  - Designed a layered system architecture with SSR/CSR rendering, REST APIs, secure internal service communication, and containerized deployment via Docker Compose.
- **Fuzzy Control of Truck-Trailer Auto Parking in the Presence of Obstacles** Dec 2013
  - Developed a hierarchical fuzzy logic controller in MATLAB/Simulink for autonomous reverse parking of a truck-trailer system.
  - Modeled the nonlinear kinematics of a 4-DOF trailer-truck vehicle and implemented steering control algorithms for reverse maneuvering.
- **Design and Programming of a 3-DoF Robot Controller Circuit Board** Jul 2013
  - Designed and developed an embedded control system for a 3-DoF robotic manipulator using an ATmega32 AVR microcontroller.
  - Implemented simultaneous position and motion control of two servo motors and one DC motor to execute predefined robotic trajectories.
- **Control System Design of a Nonlinear Inverted Pendulum System** Jun 2013
  - Developed a mathematical model of a nonlinear inverted pendulum-on-cart system using Newtonian dynamics and derived governing equations of motion.
  - Linearized the nonlinear system around the upright equilibrium point and formulated both transfer function and state-space representations.
- **Hydraulic Circuit Design for Ring Rolling Machine** Jun 2013
  - Designed hydraulic circuits for a ring rolling machine based on specified force, stroke, pressure,

and speed requirements.
  - Performed hydraulic actuator sizing calculations, including cylinder bore and piston rod diameter selection for multiple machine functions.

- **LabVIEW-Based Simultaneous Control of 3 CNC Stepper Motors** Mar 2013
  - Designed and developed a LabVIEW-based motion control system for simultaneous operation of 3 stepper motors used in CNC applications.
  - Implemented real-time motor speed control by converting user-defined rotational speed inputs into synchronized timing and pulse-delay sequences.

- **Stable Path Planning and Body Angle Analysis of a Biped Robot** Jun 2012
  - Analyzed the influence of torso/body angle parameters on the dynamic stability of a biped robot using Zero Moment Point (ZMP) stability criteria.
  - Developed kinematic and dynamic models of a multi-link biped robot, including body segments, joints, mass distribution, and inertia properties.